Venkata Lakshmi Parimala Pasupuleti

AI Architect · Internal AI Features · LLM Evaluation · Responsible AI · RAG Pipelines · Python · LangGraph · Enterprise SaaS

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SUMMARY

Product-minded AI Architect with 9+ years building enterprise line-of-business applications and two years shipping production AI features end-to-end. Built **Leasel**: three internal AI surfaces sharing one knowledge repository - document intelligence, agentic investigation, and self-service - framed the problem, prototyped the architecture, shipped to production, and operate it with CI-integrated evaluation and monitoring. Built **AgentProbe**: open-source LLM evaluation toolkit with offline evals, tracing, feedback loops, safety guardrails, and CI regression harness - the evaluation and observability infrastructure that keeps production AI features reliable and measurable over time. Designed with responsible AI practices throughout: HITL escalation, audit logging, hard safety boundaries, and PHI/access-control compliance. At Entrata (9+ years, 100K+ users), owned enterprise line-of-business SaaS modules end-to-end in a regulated multi-tenant environment - backend services, APIs, data pipelines, and cross-functional stakeholder delivery. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

AI Features · Internal Tooling | LLM Evaluation · Offline Evals | Responsible AI · Guardrails · HITL | RAG Pipelines · Agent Frameworks | LangGraph · LangChain | Tracing · Observability · Feedback Loops | Prompt Engineering · Structured Outputs | Anomaly Detection · ML Models | Python · JavaScript · TypeScript | SQL · PostgreSQL · Snowflake | Anthropic · OpenAI · Azure OpenAI | AWS · Docker · Kubernetes | CI/CD · Automated Testing | Enterprise SaaS · Line-of-Business Apps | REST APIs · System Integrations | Stakeholder Communication | GTM · Internal Business Applications | Vector Stores · Embeddings | Audit Logging · Access Control · PII · PHI

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, AI-powered features, and data pipelines end-to-end; surfacing business insights into application-facing APIs and reporting layers for stakeholders.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Built a production agentic AI workflow (LangGraph) for internal operations - framed the problem with business stakeholders, prototyped, evaluated, and shipped to production with responsible AI practices: validation guardrails, hard compliance boundaries, and structured audit logging throughout.
- Deployed MCP server connecting agent outputs to enterprise systems; engineered full RAG pipeline from document ingestion to FastAPI serving with defined API contracts.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, enterprise line-of-business)

- **Owned enterprise line-of-business modules end-to-end** for 7+ years in a regulated multi-tenant environment - backend services, APIs, data pipelines, and compliance workflows serving 100K+ users; translated complex business requirements into production software with measurable outcomes; zero compliance violations.
- **Built and maintained cross-functional stakeholder relationships** - collaborated with product, operations, compliance, and business teams to scope, prioritize, and deliver AI-adjacent automation features; communicated technical concepts and tradeoffs clearly to non-technical stakeholders and leadership.
- **Drove observability and reliability** - built internal debugging tools reducing debugging time by 40%; owned production incident response, root-cause analysis, and postmortems; maintained bug count below 10 for 6+ consecutive months across a 100K+ user platform.
- Optimized 200+ SQL queries (30% gain); PHI-encrypted AWS S3 migration (100% coverage, zero incidents); Python/Bash automation eliminated 80% of manual workload; led microservices redesign (Docker) - 50% scalability improvement; mentored 5+ engineers; 300+ code reviews.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built line-of-business modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration; delivered large-scale data import and reconciliation workflows (1.6M records).

PROJECTS

Leasel - Internal AI Platform (3 Surfaces, Responsible AI Throughout) 2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Three internal AI surfaces sharing one knowledge repository: **document intelligence** (OCR, RAG pipelines, hybrid search, citation grounding), **agentic investigation** (8-agent LangGraph pipeline with tool-use and HITL escalation), and **self-service** with hard safety boundaries - regulated decisions always route to staff, enforced in code. Responsible AI built in: audit logging on every agent decision, access-controlled retrieval, no AI auto-resolution of compliance or financial outcomes.
- CI regression gate; 93.6% precision and 87.9% F1 across 1,240 test cases (10 compliance rules, 5 US states + federal).

AgentProbe - LLM Evaluation, Tracing & Observability Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLM evaluation toolkit: **offline evals** with golden datasets and LLM-as-judge scoring; **tracing** and structured logging across agent runs; **feedback loops** via CI-integrated regression harness that catches degradations before they reach production; 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance); multi-provider routing (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). The evaluation infrastructure that keeps AI features measurable and reliable over time.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017